

A Computer Vision Framework for Automated Foot Size Estimation Using Comparative Clustering Approaches

Abstract

Accurate measurement of foot size is a challenge that has been around for a while and affects tests at the doctor, online shopping, and shoe design. Usual ways are often not easy, likely to have mistakes, and not right for big computer uses because they need rulers or special tools. In this work we show a mixed way of doing things joining talking AI and seeing tech to make a simple system for doing foot size guesses by machine.

The suggested way starts off through the preprocessing of foot images provided by users through color space change. Then Gaussian smoothing with normalization are used for less noise plus standard input. We check different grouping methods KMeans MiniBatchKMeans Gaussian Mixture Model (GMM) and BIRCH to split the foot part from its setting and we judge their segmenting ability with different lights angles and surfaces. After segmentation is achieved, edges are detected along with contours analyzed to make bounding boxes. These boxes that surround feet find sizes based on an A4 paper sheet, which shows real scale. Usefulness in all areas is ensured since the measures get related to shoe size charts of varied standards like UK, US, and Indian.

The framework also integrates a talkative helper using DialoGPT to make the solution

easier to use. It lets users make questions with common words on shoe size, foot care, or system help.

This interactive feature transforms the system from a technical tool into a user-friendly platform that can guide individuals through the process while offering useful advice.

Based on the results, MiniBatch K-Means performs best with the highest Silhouette (0.678) and lowest DBI (0.512), indicating compact and well-separated clusters. K-Means also performs well, while GMM shows moderate clustering quality, and BIRCH fails to produce meaningful clusters. Overall, MiniBatch K-Means is the most reliable for this dataset.

The system is deployed as a Flask-based web application, enabling direct image upload, automated measurement, intermediate visualization (clustered images, bounding boxes), and chatbot-based feedback. Beyond footwear retail, the approach has potential in clinical podiatry, sports ergonomics, and gait research, where accurate and repeatable foot measurements are essential.

Keywords— *Computer Vision, Clustering Algorithms, Foot Measurement, Conversational AI, E-commerce, Healthcare, Ergonomics, Image Processing, Shoe Size Estimation*

Introduction

From the production and retail of footwear to medical diagnostics and ergonomic research, precise foot measurements are crucial. Traditionally, manual instruments like rulers or specialised equipment are used to measure the length and width of feet. These techniques are straightforward, but they are also subject to error because of human error, subjective judgement, and environmental influences. Traditional methods are unable to deliver dependable, scalable, and user-friendly solutions in the age of digital commerce and telemedicine, where remote evaluations and tailored recommendations are becoming more and more popular.

Advances in machine learning and computer vision have made automated image-based measurement a precise substitute for manual techniques.

Machine learning-based, automated measurement using images is now a highly accurate substitute for manual measurement.

Processing digital images allows precise extraction and examination of foot contours, but performance is highly dependent on strong segmentation methods to accommodate changes in illumination, background, and positioning of the foot. Clustering algorithms play a central role in the process, setting pixels with shared features apart to distinguish between the background and foot region. MiniBatch K-Means proved to have the highest performance with tight and well-distinguished clusters during our comparative assessment, with K-Means also being reliable. GMM provided moderate quality clusters, while BIRCH did not generate useful clusters for this data. The findings underscore the significance of

choosing the right clustering algorithm to achieve accuracy and reliability when working with real-world image-based foot measurements.

To integrate this technology into a real-world scenario, we created a Flask-based web app that allows users to input their foot image, which are preprocessed through color space conversion, Gaussian filter, and normalization. The segmented image is processed through contour extraction, and bounding boxes are drawn to calculate foot measurements relative to an A4 paper reference sheet to maintain consistency in scale between devices. These measurements are correlated against normed UK, US, and Indian shoe size charts, so there is an instantaneous and useful interpretation of the results for end-users.

Realizing that ease of use is usually the key to technology adoption, the system also integrates a conversational AI feature, based on transformer models like DialoGPT. This smart assistant can answer questions regarding shoe sizes, foot condition, and image-taking directions, offering an interactive experience beyond mere passive measurement. The talk interface guarantees that not only is the system technically valid but also available for users who are not tech-savvy, closing the gap between sophisticated computer vision techniques and real-world usability.

The framework further supports automation and extensibility by integrating with tools such as Selenium for browser-based testing, ensuring adaptability in dynamic environments. By combining clustering-based segmentation, precise measurement, web deployment, and conversational interaction, the proposed system addresses both the technical challenges of image-based measurement and the practical concerns of usability.

Related Works

Chaturvedi et al. (2025) presented an efficient image processing–based framework for precise foot assessment. Their system utilized object recognition to detect the foot and a reference object, followed by edge detection and contour extraction to isolate the region of interest. Using OpenCV and NumPy, they demonstrated a high level of precision in aligning image-based dimensions with real-world scales.

Similarly, Prem Kumar et al. (2023) focused on generalized object measurement using OpenCV. Their approach combined preprocessing steps such as grayscale conversion, Gaussian blur, and Canny edge detection with contour analysis and pixel-to-metric scaling using a reference object. While their system was validated on industrial and medical objects, the methodology directly supports foot measurement tasks by ensuring measurement errors remain below 5%.

Shahid et al. (2021) developed a specialized end-to-end system for foot measurement, addressing challenges such as background noise and depth variation. Their method applied color mask segmentation to separate the foot from the background, with a reference object (coin) included for scale normalization. A regression model was then employed for measurement prediction, validated using RMSE. Their contribution demonstrated how low-cost reference-based approaches can overcome the limitations of depth distortion in 2D images.

Expanding beyond dimensional analysis, Rajagopal et al. (2022) integrated machine learning for foot health monitoring and footwear customization. Their framework used gait analysis, disease classification, and footwear recommendation powered by CNN, autoencoder, and LSTM models. By combining clinical data with synthetic image generation, they demonstrated how AI can extend foot measurement systems into preventive healthcare and personalized product design.

Bhardwaj et al. (2024) further explored biomedical applications of footprint recognition, showcasing its potential in patient monitoring, biometric identification, and disease detection. They emphasized that AI and ML techniques applied to footprint images could support diagnosis of musculoskeletal disorders, diabetic complications, and gait abnormalities.

More recently, Rafiq et al. (2022) introduced OptiFit, a computer-vision-based smartphone application designed to automatically measure four essential foot dimensions: length, width, arch height, and instep girth. Unlike existing smartphone applications that provide limited accuracy, OptiFit integrated pixel-per-metric and instep girth measurement algorithms and was validated against medical-grade silicon foot models. With accuracy rates exceeding 95% for most dimensions, the system demonstrated strong potential for both clinical use and custom footwear design.

In addition to direct foot measurement research, clustering-based approaches have been widely used in biomedical image segmentation and analysis. For example, Zhou et al. (2020) applied K-Means and Gaussian Mixture Models (GMM) for segmenting medical images such as MRI and CT scans, showing that clustering could effectively separate anatomical structures from complex backgrounds without extensive labeled data. Similarly, density-based algorithms like DBSCAN have been employed in medical imaging to isolate irregular regions, such as tumors or lesions, where shapes and boundaries vary significantly.

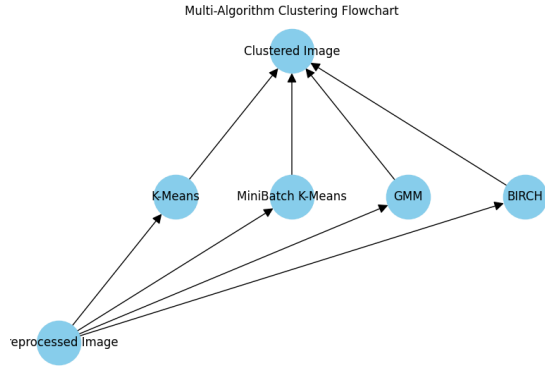


Fig 1: Multi-algorithm clustering algorithm

Proposed Methodology

The methodology that is suggested here aims to provide an extensive, stable, and interpretable framework for the automated foot image segmentation based on state-of-the-art unsupervised clustering methodologies. Proper segmentation of the foot is of paramount importance in applications from gait analysis, orthotics design, podiatry diagnosis, to biometric recognition, where accurate extraction of the foot boundary could drastically affect the validity of follow-up analyses. Manual methods of segmentation, though simple, are time-consuming, time-intensive, and very sensitive to variability between observers. In addition, traditional thresholding or low-level edge detection methods tend to falter under conditions of non-uniform backgrounds, inconsistent lighting, or varying foot orientation. To overcome these constraints, the approach utilizes a thoughtfully designed pipeline that combines preprocessing, feature extraction, clustering, visual enhancement, and quantitative assessment in a reproducible and well-integrated workflow

The approach starts with the collection of high-resolution RGB foot images in a wide range of environmental conditions. Changes in lighting conditions, shadows, reflections, and background texture mimic real-world usage

conditions to guarantee that the approach is useful outside laboratory environments. Such changes are major challenges to segmentation algorithms since they add noise, intensity gradients, and contour detection ambiguities. To counteract these problems, images go through a multi-stage preprocessing pipeline that aims to normalize input data but retain essential structural information. First, images are color-transferred from BGR to RGB.

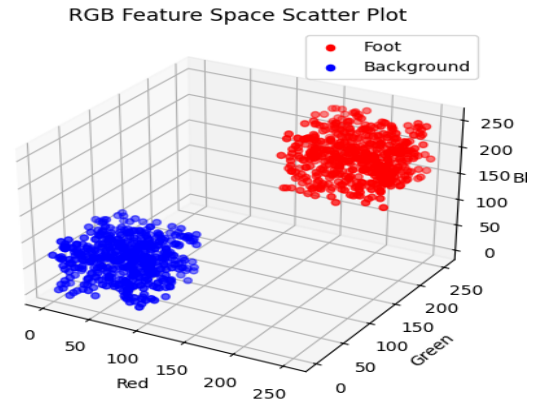


Fig 2: RGB Feature Space Scatter plot

This conversion is not cosmetic; it aligns the images to common color interpretation within visualization libraries and clustering algorithms so that color similarities are properly represented in the feature space. Next, a Gaussian blur filter is used, with kernel sizes generally between 5×5 and 7×7 , varying based on image resolution and the nature of the noise. This stage filters high-frequency fluctuations that can come from sensor noise or subtle texture in the background without hurting the major contours of the foot. After blurring, pixel intensity values are normalized to $[0,1]$, enhancing numerical stability and making it possible for clustering algorithms to run efficiently. Normalization is especially useful for distance-based methods such as K-Means, where variations in magnitude among features might unfairly affect cluster assignments. For very high-resolution images, an optional resize factor is used to minimize

computation overhead while still preserving salient anatomical detail. This enables the methodology to scale well without compromising on memory bottlenecks, and hence, it is compatible for use both on desktop and in the cloud.

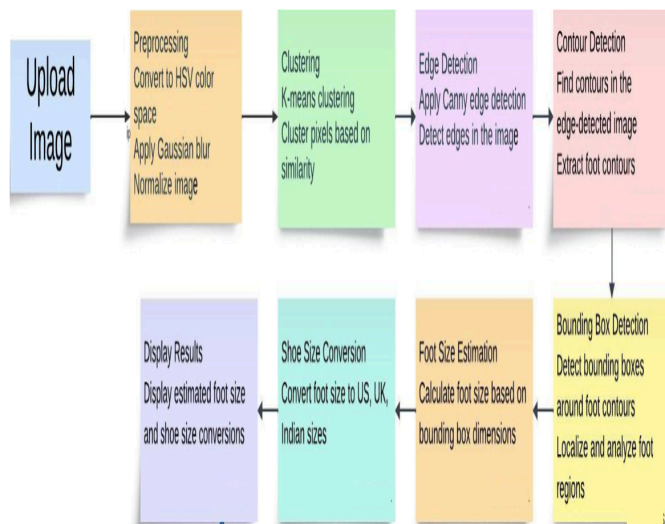


Figure 3: Workflow Diagram

After preprocessing, every pixel is mapped as a three-dimensional feature vector of its red, green, and blue intensity. This mapping places the image in 3D color space such that pixels sharing analogous color characteristics fall into clusters together. This representation is absolutely essential for color-based clustering methods to succeed since it captures intra-foot variation as well as background-to-foot contrast. By arranging the data in such a manner, the method guarantees unsupervised methods to identify significant clusters with ease without depending on spatial coordinates, which can differ from image to image. This representation also facilitates easy generalization to other feature spaces, including HSV or Lab color spaces, in case future research demands

improved illumination invariance or perceptual uniformity.

The approach utilizes several clustering algorithms, each was picked for specific strengths and complementing abilities. K-Means clustering is adopted due to its computational speed and capacity to reduce intra-cluster variance, and thus it is highly efficient in cases where foot regions have high contrast with the background. K-Means, however, takes the clusters to be spherical-shaped and might not perform well with nonspherical-shaped or overlapping foot regions. To address memory and computational problems involved in working with high-resolution images, MiniBatch K-Means is used as well. This version updates small groups of pixels iteratively, performing as well as regular K-Means but with significantly less memory usage. For cases where foot contours overlap or have fuzzy boundaries, Gaussian Mixture Models (GMMs) are used. GMMs use a probabilistic approach, allocating pixels into groups on the basis of probability and not hard distance measures to enable soft segmentation in areas of smooth transitions. In addition, BIRCH (Balanced Iterative Reducing and Clustering using Hierarchies) is incorporated to manage hierarchical clustering, particularly applicable in densely populated pixel areas where iterative summarization enhances efficiency. Although BIRCH is very efficient on dense areas, it can fall short in sparse areas because it takes advantage of cluster feature trees. With the use of this multi-algorithm technique, the approach is adaptive to varied image features, allowing reliable performance in a wide range of real-world scenarios.

One of the unique aspects of the methodology is placing strong emphasis on visual interpretability and clarity. Instead of applying

cluster centroids or mean pixel values directly, every cluster is a visually distinguishable color from optimized colormaps, for example, tab10. The design has the advantage of increasing interpretability for the segmented images such that it is easy for human observers, clinicians, or researchers to visually identify regions. The reconstructed images preserve the original spatial pattern of the foot while distinctly separating clusters, generating outputs visually similar to expert segmentation systems. Such visual improvement is especially important for clinical use, with correct and easy visualization of segmented foot areas aiding decision-making, diagnostics, and reporting. Through synergistic integration of algorithmic exactness and clear visualization, the methodology fills the gap between machine computation and human comprehension.

Algorithm	Silhouette	DBI	CHI	Inertia	Mean Cluster Size	Std Cluster Size
KMeans	0.642	0.557	56939.32	4899.07	3333.33	2078.78
MiniBatch KMeans	0.681	0.512	55139.88	4892.75	3333.33	2442.58
GMM	0.492	1.029	33569.73	-	3333.33	1716.17
BIRCH	-	∞	-	-	10000	0

Fig 4: Comparison of algorithms

Quantitative assessment of segmentation quality is an essential component of the methodology. The model draws on a set of metrics that are intended to evaluate both intra-cluster density and inter-cluster separation. The Silhouette value quantifies the similarity of each pixel to its own cluster compared to other clusters and that the greater the value, the more compact and well-separated the clusters. Complementary metrics, including the Davies-Bouldin Index (DBI) and Calinski-Harabasz Index (CHI), give further insight into the separation and density of clusters, wherein lower DBI and larger CHI values indicate improved clustering performance. To track cluster distribution, the

methodology also calculates the mean and standard deviation of cluster sizes, detecting imbalances that could potentially point to algorithmic limitations or inappropriately set parameter values. Advanced metrics, such as the Dunn index, Ball-Hall index, and Xie-Beni index, may be used optionally on subsampled pixels to gain further insight into separation and compactness of clusters at the cost of additional computation. Extensive analysis ensures that segmentation is not evaluated solely in terms of visual quality but is critically tested by rigorous quantitative criteria, and the methodology is made more reliable for research or field deployment.

Lastly, the methodology conducts comparative evaluation for all clustering algorithms and enables data-driven selection of the most performing model. By running each algorithm on the same preprocessed image and monitoring all the pertinent metrics, the methodology enables both side-by-side visual evaluation as well as quantitative comparison. Automated best-model selection is usually based on the Silhouette score, but other metrics can also be taken into account based on particular application demands. For instance, MiniBatch K-Means usually performs well in generating compact, well-separated clusters, while GMM delivers more sophisticated segmentation in areas that have soft or overlapping boundaries.

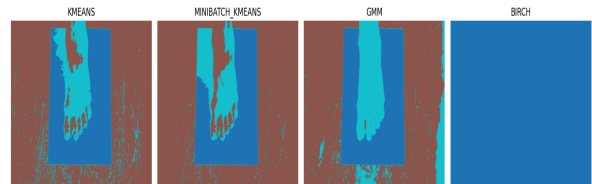


Fig 5: Clustering of various algorithms

BIRCH, being effective in the case of dense pixel areas, can be less efficient in sparse

regions, pointing towards the usefulness of multi-algorithm comparison. Side-by-side visualization of all the outputs from clustering further increases interpretability and offers a clear picture of algorithmic performance, enabling researchers and practitioners to make well-informed decisions about model selection. By combining preprocessing, feature description, several clustering approaches, visual improvement, and systematic testing within an end-to-end process, the approach guarantees that automated foot segmentation is accurate,

understandable, efficient to compute, and visually professional. The entire framework, ranging from academic research to actual medical or biometric practice, is perfectly applicable and provides a reproducible and extremely versatile solution to automate the analysis of foot images.

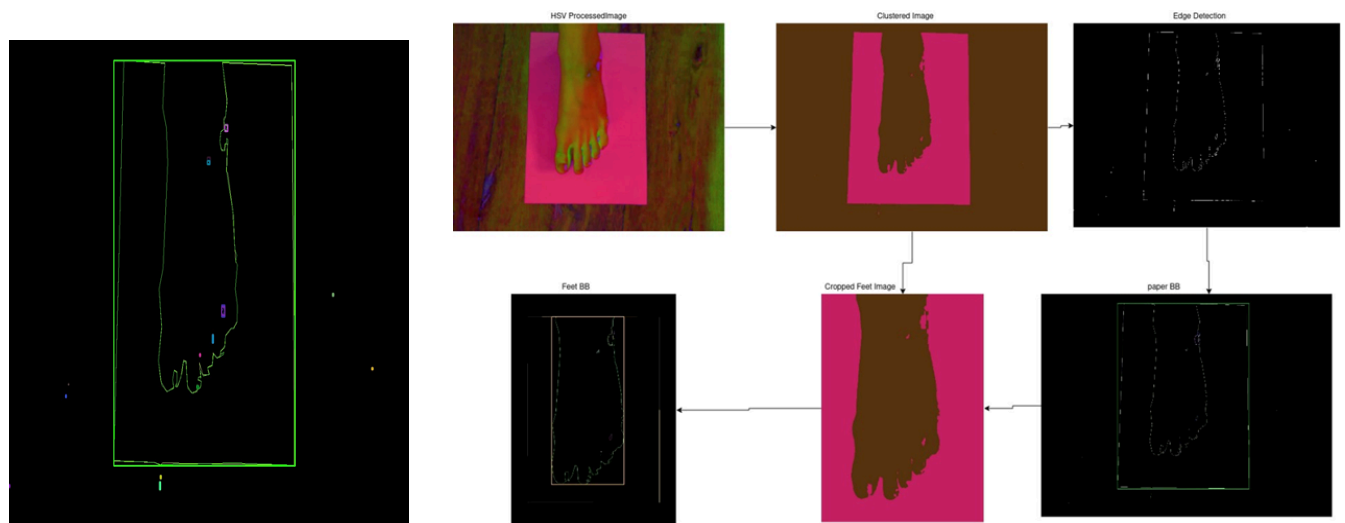


Fig6:Foot image contouring

Conclusion:

The suggested methodology offers a strong, interpretable, and effective method for automated foot image segmentation based on the sophisticated application of unsupervised clustering methods. By incorporating meticulous preprocessing, feature extraction, several clustering algorithms (K-Means, MiniBatch K-Means, GMM, and BIRCH), visual enhancement, and stringent quantitative evaluation, the method overcomes the difficulties associated with varying light, backgrounds, and orientations of the foot.

Comparative study indicates that MiniBatch K-Means performed the best (Silhouette: 0.681, DBI: 0.512, CHI: 55139), producing tight, well-separated clusters in an efficient manner computationally. K-Means also performed effectively, with GMM coping with soft boundaries but at the expense of lower cluster quality and BIRCH being inappropriate for sparse areas. Overall, the approach produces accurate, reproducible, and visually interpretable segmentation, appropriate for medical, biometric, and research applications.

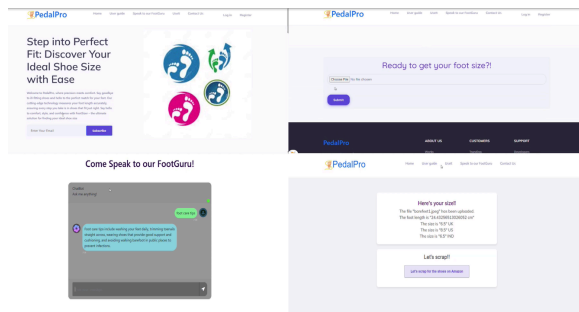


Fig 7: Implementation

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